Venu Thatikonda

Ringstraße 39, Heidelberg
+49(0) 152 15115030
thatikonda92@gmail.com
itsvenu.github.io

Education

German Cancer Research Center — **Heidelberg, Germany**
PhD student in Bioinformatics — *Oct 2016 - July, 2020*
Major focus: Large-scale sequencing projects for biomedical applications

National Institute of Pharmaceutical Education and Research — **Kolkata, India**
MS in Pharmacoinformatics — *Aug 2013 - June 2015*
Major subjects: Bioinformatics, Statistics, Molecular biology, Programming

Kakatiya University — **Warangal, India**
Bachelors in Pharmacy — *Aug 2009 - June 2013*
Major subjects: Molecular biology, Statistics, Pharmacology, Medicinal chemistry

Skills

- R statistical programming language (Bioconductor, tidyverse & dataviz tools)
- Python (pandas, numpy, scikit-learn, tensorflow & dataviz tools)
- Single-cell and bulk sequencing analysis
- Integrative analysis of multi-omics data
- Jupyter notebooks and Rmarkdown
- 'git' version control system
- bash/shell programming
- Intuitive visualization of large scale datasets
- Implementing machine learning models and inferring associations
- Ability to work on linux with high-performance computing clusters & GPUs
- Large-scale data management by writing workflows (e.g. Snakemake)
- Extracting data from public resources such as TCGA, GEO, SRA, EGA
- Software development
- Coordinating with wet-lab researchers to design experiments
- Open source data science projects (github.com/itsvenu) and blogging (itsvenu.github.io)

Work experience

German Cancer Research Center, Heidelberg — **Oct, 2016 - present**

- **PLEASE DM FOR ADDITIONAL DETAILS**
- Biomarker identification using single-cell transcriptomics and bulk multi-omics data
- Identification of drug activity mechanisms and effects in cancer cells
- Pan-cancer analysis of whole genomes and whole exomes

National Institute of Biomedical Genomics, India — **Aug, 2015 - Sept, 2016**

- Whole-exome sequencing analysis of head and neck squamous cell carcinoma patients to identify driver

gene mutations
- Characterization of heterogeneity within cancer associated fibroblasts in oral cancers by transcriptomic analysis

Research Interests

- Single-cell biology
- Bridging the gaps in understanding complex diseases such as cancer
- Machine learning in single-cell and multi-omics
- Understanding drug mechanisms in disease

Publications

- Reduced binding of MYC to chromatin is a key effect of HDAC inhibition in MYC amplified Medulloblastoma [*under review*]
- Pan-cancer analysis of mutational signatures in pediatric primary and relapsed tumors [*under preparation*]
- ALPS: Analysis routines for epigenomics data, *Bioconductor, 2019*
- Aberrant ERBB4-SRC signaling as a hallmark of group 4 medulloblastoma revealed by integrative phosphoproteomic profiling. *Cancer cell, 2018*
- A subtype of cancer-associated fibroblasts with lower expression of alpha-smooth muscle actin suppresses stemness through BMP4 in oral carcinoma. *Oncogenesis, 2018*
- Other publications Google scholar profile

Awards & Achievements

- Travel grant to attend International PhD Student Conference on Cancer, The Netherlands, 2019
- Funding to pursue PhD from ICGC-data mining project, Germany, 2016
- Gold medal from National Institute of Pharmaceutical Education and Research, India, 2015
- Funding to pursue MS from Human Resource and Development, India, 2013-2015
- Funding to pursue Bachelor of Pharmacy, India, 2009-2013
- Scholarship from LIC Golden Jubilee project, India, 2009-2013

General interests

- Writing (itsvenu.github.io)
- Open source project building
- Reading books
- Badminton